Embodied AI Through Cloud-Fog Computing: A Framework for Everywhere Intelligence

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***Abstract*—Embodied AI represents a crucial step towards achieving Artificial General Intelligence (AGI). The next paradigm of Embodied AI involves physical embodiment, enhanced perception capabilities, and adaptive automation. This advances the field significantly, paving the way for broader expansion. Despite the significant progress, existing computing frameworks, like local computation or cloud computing, struggle to meet the substantial demands of Embodied AI. The Cloud-Fog Embodied framework, namely based on CFA (cloud-fog automation) offers a promising solution to address these challenges. Our goal is to drive integration across multiple domains, including AI, robotics and industrial production, to tackle multifaceted challenges and seize opportunities to achieve AGI in the future.**

***Index Terms*—Embodied AI, agent, cloud-fog automation, pretrained foundation models**

I. Introduction

Following the rise of Large Language Models (LLMs), the discussion around Artificial General Intelligence (AGI) has once again sparked interdisciplinary debates [1]. Leveraging the capabilities of pretrained foundation models represented by LLMs has made it possible for AGI to further advance. The key to achieving AGI lies in the emergence of a more advanced agent. Within the field of robotics, the concept of Embodiment has been repeatedly mentioned [2], imbuing a new significance in the context of recent research as referring to agents with physical embodiments. Over time, the concept of Embodied AI, based on the notion of Embodiment, has been cited and explored by industry giants such as NVIDIA, Meta, Microsoft, and Tesla. In the subsequent section of this paper, we define and explore the transitional research stages leading up to the deployment of Embodied AI in real-world scenarios.

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However, the journey towards fully realizing Embodied AI is not without its challenges. The requirements for this are detailed in the third section of the paper. First, these requirements are constrained by the existing cloud and endpoint architectures, a topic that is further explored in the fourth section. Second, to address these challenges, a promising solution is provided by the Cloud-Fog Embodied framework based on Cloud-Fog Automation (CFA). This framework, introduced in the fifth section of the paper, offers a promising solution with significant advancements in flexibility, adaptability, and privacy compared to previous structures.

II. Evolution of Embodied AI

Embodied AI is a multifaceted concept that bridges various disciplines. It consistently garners notable interest whenever AI sparks a new technological revolution. Embodied AI integrates cutting-edge disciplines to pursue pathways toward achieving Artificial General Intelligence (AGI) from diverse fields. The boom of pretrained foundation models and their astonishing capabilities marked a significant transition. Embodied AI began to move from passively solving structured tasks to becoming a dynamic agent. It is now capable of autonomous planning in diverse and complex environments [3]. Within this innovative paradigm, Embodied AI can learn and adapt in a physical environment without continuous human intervention [4]. Agents based on Embodied AI, through receiving external stimuli, engaging in self-directed interactions, and autonomous learning processes, mirror human-like capabilities. After going through these processes, it will eventually achieve self-iteration and evolution. These capabilities align with our requirements for adaptability and time series in Embodied AI, as illustrated in Fig 1.

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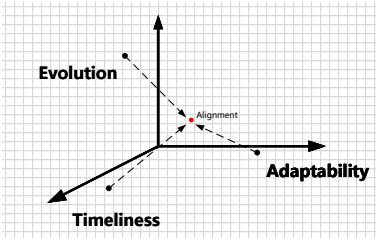


Fig. 1. Embodied AI inherently possesses the subjective adaptive capability, which is directly proportional to its own evolutionary level. Both of these capabilities are highly aligned with the time series data, marking a prominent characteristic of Embodied AI.

Before deploying Embodied AI into the real world, there are several transitional research stages that need to be addressed [5]. Habitat [6], researchers construct 3D environments to test whether Embodied AI can perform the tasks assigned to agents. Since the emergence of Large Language Models, many new research approaches have surfaced. Cooking robots leverage the capabilities of pretrained foundation models to effectively control complex scenarios like cooking that unfold over time [7]. By utilising multimodal pretrained foundation models with efficient visual processing and semantic understanding capabilities, the costs of learning and training scenarios are reduced [8]. Spatialvlm [9],designing 3D frameworks based on pretrained foundation models upgrades the modalities through which agents process information. Additionally, developing practical frameworks for deploying pretrained foundation models on low-performance platforms lowers the deployment complexity of agents [10].

From the above work, we conclude the key aspects of Embodied AI include:

- **Physical Embodiment and Simulation Integration**
  Embodied AI systems are furnished with physical robotic entities or platforms that possess the capability to sense and interact with the tangible world, rather than solely existing in the virtual domain [11]. Embodied AI also possesses a disembodied AI brain, which can operate in virtual simulations as well.
- **Incorporating Multi-Modal Sensory Perception**
  Embodied AI systems utilise multiple sensory modalities like vision, touch, and audition to perceive and comprehend their surroundings, mirroring how humans experience the world. With the assistance of time series data, perception involves aligning different modalities with each other [12].
- **Robust Generalization Capabilities**
  Facing physical environments where not all scenarios have been previously encountered during training, Embodied AI learns to grasp the basic elements of different environments and the rules governing system behaviours [13]. By observing the interplay between these simple elements and rules, Embodied AI extends the emergence of complex behaviours and enables the manifestation of matching behaviours under new environments and rules of system behaviour.
- **Adaptive Automation**
  Given the diverse and intricate nature of scenarios targeted by Embodied AI, it necessitates adaptive capabilities. In varying situations, Embodied AI must exhibit high real-time autonomous decision-making abilities while balancing reliability and safety considerations.

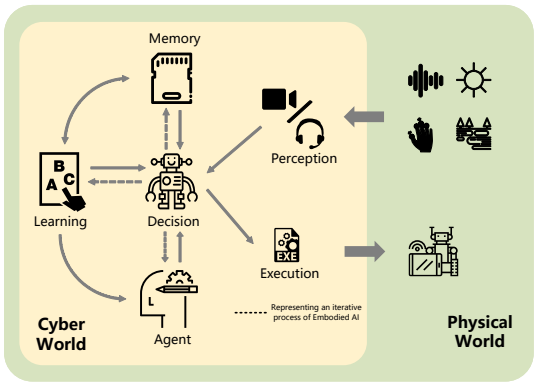


Fig. 2. A operational paradigm of Embodied AI in various scenarios involves the utilization of perception-driven environmental information, through the orchestration of Learning, Memory, and Agent modules under the guidance of Embodied AI. This integration enables a closed-loop system where data flow and decision-making processes are seamlessly executed.

In the process of investigating the development of Embodied AI, inspired by pretrained foundation models, we propose that Embodied AI will evolve into a generalised concept in the future. Embodied AI is not limited to humanoid robots at the forefront of current technology; it encompasses any agent capable of perceiving external information, utilizing this information for autonomous cognition, decision-making, learning, and action. Fig 2 represents our envisioned paradigm of Embodied AI. For instance, in industrial production, robotic arms on assembly lines with Embodied AI capabilities can utilize sensors such as vision and ultrasonic sensors to assess the status of the assembly line. They can autonomously make decisions to match the overall production situation of the assembly line, adjust operation speeds, correct angles automatically, and so forth.

III. The Demand for Computing Horsepower and Control

The journey towards achieving genuine Embodied AI is still far from reach. An analysis of current research projects reveals that Embodied AI imposes substantial computing horsepower and control demands. Ongoing Embodied AI projects involve environmental perception tasks, with specific functionalities and fine-tuning tasks based on pretrained foundation models, all reliant on GPU clusters. The integration of Embodied capabilities into physical robotic platforms necessitates significantly more resources compared to conventional disembodied AI methods [14]. For instance, embedding computing modules in the form of soft, deformable materials within the body of robotics. However, this involves innovations in soft electronics. [15], [16]. Therefore, currently, finding a balance in the complexity and adaptability of Embodied AI projects involves

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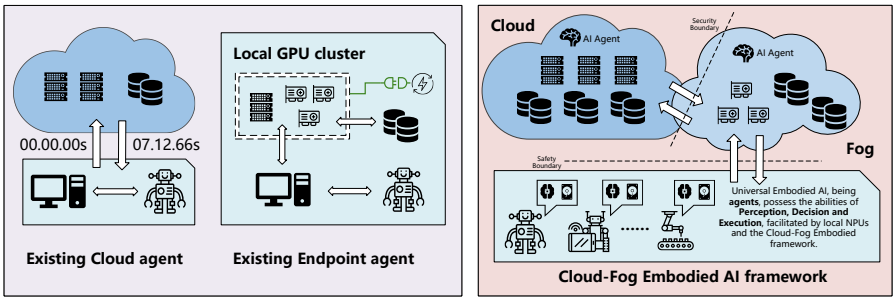


Fig. 3. In the architecture of Cloud and endpoint, Embodied AI struggles to achieve a balance between real-time control and computing horsepower. However, in the Cloud-Fog Embodied framework, computing horsepower can freely flow and be efficiently allocated by AI. The Cloud and Fog ends possess flexibility and scalability, while endpoint hardware is equipped with chips that have a certain level of computational capability to execute simple tasks. Supported by relevant algorithms on the software end, this setup ensures both privacy protection and adaptability.

a trade-off between the control system's complexity and the adaptability of the Embodied AI agent. Similar to leveraging the advantages of low-cost communication, adopting a cloud-endpoint architecture reduces the local demand for resources.

IV. Limitations of Cloud-Endpoint Architecture

Currently, Embodied AI operates on a cloud and endpoint separation architecture. In this setup, projects resembling Embodied AI extensively use API calls to complete operations. The capabilities represented by APIs originate from the cloud. Communication between the endpoint and cloud is facilitated through data transmission. Cloud-based models and processing capabilities are typically separated from physical agents running at the edge. This separation may introduce latency issues as agents need to continuously transmit data to the cloud and receive responses.

In an endpoint deployment mode for intelligent agents, the entire module must be supported by local GPUs, ranging from single graphics cards to GPU cluster servers [17]. When executing operations in Embodied AI, the physical size and weight of GPUs present design challenges for mobile or spatially constrained operations. Additionally, the high cost associated with configuring GPUs and relevant infrastructure cannot be overlooked.

Furthermore, this architecture lacks autonomy and adaptability. Heavy reliance on cloud-based resources may diminish an agent's capacity for independent decision-making and real-time adaptation to evolving conditions. Local computing power may not facilitate self-learning, self-iteration, and the evolution of Embodied AI. These bottlenecks constrain the further advancement of Embodied AI.

V. Cloud-Fog Embodied framework

We propose the Cloud-Fog Embodied framework (CFEF) based on the Cloud-Fog Automation framework [18]–[20], as illustrated in Fig 3. The cloud offers large-scale capabilities. The internal structure of the cloud is illustrated in Fig 4. The internal structure of the fog is similar to that of the cloud. However, the fog provides more specific capabilities compared to the cloud, offering comprehensive support tailored to specific projects.

- **Computational Resources and Scalability**
  CFEF utilizes the synergy of cloud computing and fog/endpoint computing. It provides ample computational power for Embodied AI. This distributed computing architecture offloads resource-intensive tasks to the cloud. Meanwhile, it enables real-time processing and decision-making at the fog, close to the physical Embodied AI agents. CFEF can reduce latency by bringing computing and data storage physically closer to the generation location [21] Additionally, computational resources in the cloud and fog can scale independently without impacting the operation of Embodied AI.
- **Flexibility and Adaptability**
  CFEF plays a key role in coordinating the productionThe virtualization and convergence of computing horsepower and communication networks enabled by CFEF allows for the flexible deployment of Embodied AI use cases across various environments and applications. This flexibility supports the adaptability of Embodied AI systems to different operational contexts and changing requirements, enabling autonomous evolution of Embodied AI as computing and networking resources can be dynamically allocated and reconfigured.
- **Interoperability and Integration**
  CFEF approach enables the seamless integration of Embodied AI systems with other industrial automation technologies, facilitating smooth data exchange, communication, and collaboration among diverse components. This interoperability plays a critical role in the widespread adoption and deployment of Embodied AI. Particularly, as the form of Embodied AI evolves towards a more general direction, it is inevitable that Embodied AI will be deployed in different scenarios. In increasingly complex real-world environments, Embodied AI must interact with a variety of systems and devices to fulfill its functions effectively.
- **Security and Privacy**
  For endpoint devices that possess a certain level of computing horsepower, simple decision-making can be performed locally, avoiding the need to upload data to the cloud. Both the fog and endpoint, as well as the cloud and fog, incorporate various isolation measures to ensure data security. Local endpoint deployment of databases can enhance security by allowing users to store their data

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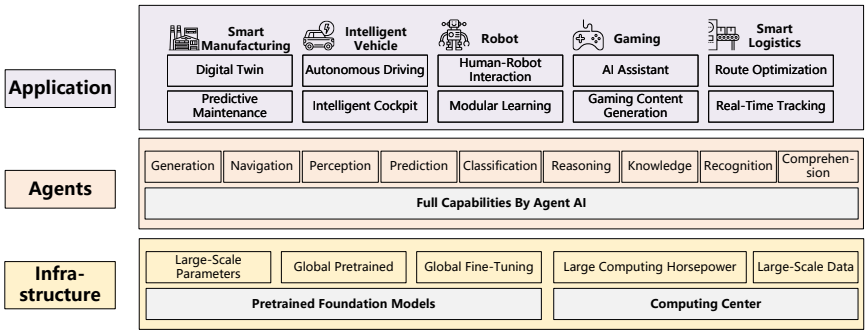


Fig. 4. The internal structure of the cloud consists of several components. The infrastructure component includes computing horsepower capabilities and large-scale data storage functions. The agent component is built upon the foundational abilities of various pretrained foundation models. The application component provides specific tasks that can be invoked for downstream tasks.

locally [22]. CFEF supports related algorithms such as federated learning, which enables Embodied AI agents to collaboratively train machine learning models without sharing their raw data [23], addressing privacy concerns.

VI. Conclusion

In the paper, we mentioned the key role of Embodied AI in implementing AGI and summarized its current definition. We presented the development status of existing projects related to Embodied AI, analyzed the requirements of Embodied AI from these projects, and summarized the existing development frameworks of Embodied AI. Building upon this research, we proposed a comprehensive theoretical framework based on Cloud-Fog Automation, providing a broad theoretical background for our discussion. This paper aims to offer a new perspective for ongoing and future Embodied AI research, fostering its development in domains.

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